The Title of Gaduation Project

CSE 496
First Presentation

Ahmet Bitircek

Advisor: Asst. Prof. Alp Arslan Bayrakci
September 2025

Contents

Project Definition

We build a **training-free, explainable system** that watches a production / assembly line and decides whether each video clip is **normal or anomalous — and explains why.**

Objective

Detect line anomalies (wrong colour, orientation, missing part, belt stop, motion) without collecting or labelling any training data.

Approach

Prompt a frozen Vision-Language Model (qwen3.6-plus) with the scene description, anomaly rules and learned guiding questions.

Output

A clip-level verdict — normal vs anomaly — plus the anomaly type and a short, human-readable reason for the decision.

Key constraint

Zero training, zero fine-tuning. The model adapts only through the prompt, so it re-targets to a new product via natural language.

Why classic computer vision struggles on the factory floor

GEBZE TECHNICAL UNIVERSITY

Data hunger

Classic detectors need thousands of labelled defect images. In an optimised factory, real defects are rare — collecting them is slow and costly.

Retraining cost

Every product or layout change forces a full re-train. We want a system that adapts from a natural-language spec, not new training data.

Black box

Standard models flag an anomaly but cannot say WHY. Operators need a human-readable reason before they trust and act on an alarm.

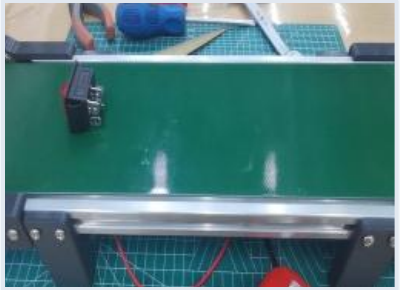
OUR APPROACH — ZERO-SHOT PROMPTING

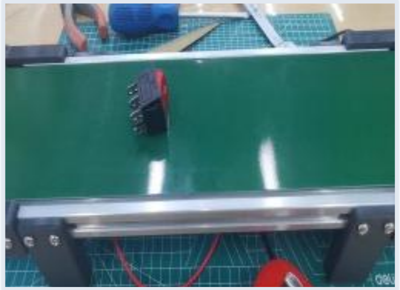
Instead of training a detector, we prompt a **frozen Vision-Language Model (qwen3.6-plus)** with the scene description and the anomaly rules. Zero training, full explainability.

From a public benchmark to a dataset we can trust

PHASE 1 · STARTING POINT

IPAD benchmark — button conveyor





- R01 push-button & S08 sorting scenes
- Anomalies: pin angle, cap colour, sorting / clog
- **Only ~256×256 — too coarse for subtle defects**

too low-res to judge colour & angle

PHASE 2 · OUR OWN DATASET

Custom pen-conveyor rig

- 13 clips · recorded in 4K → 720×1280 for the API
- 7 controlled anomaly types · 3 identical pens · fixed camera
- Full control over what is normal vs anomalous

Seven controlled anomaly types on the pen line

NORMAL

Three identical pens, caps aligned, belt moving

WRONG COLOUR

One pen differs in colour from the others

WRONG ORIENTATION

A pen points the wrong way on the board

MISSING CAP

A pen is missing its cap

MISSING PEN

Fewer than three pens are present

TEMPORAL / MOTION

Pens shift mid-clip while belt looks normal

Belt-stop / freeze and a combined (orientation + missing-cap) case complete the 13 clips.

Sample clips from the pen-conveyor rig

Recorded in 4K, downscaled for the API. **Click a clip to play it.** Same fixed camera, three pens — only the anomaly changes.

VERA: Verbalized Learning for VAD (CVPR 2025)

Ye, Liu & He — *Explainable VAD via Verbalized Learning of VLMs.* Insight: a vague prompt gives only 53–65% AUC; concrete, **learned guiding questions** do far better — with the model frozen.

1 · Frozen VLM

No fine-tuning, no extra reasoning module. The model is adapted only through the prompt.

2 · Verbalized learning

Questions are learnable: a learner answers, an optimiser rewrites them from mistakes — using only coarse labels.

3 · Coarse-to-fine

Segment scores → scene-context ensemble → frame scores via temporal smoothing. SOTA on UCF-Crime & XD-Violence.

→ We keep VERA's learner/optimiser question-learning core, but simplify it to **clip-level verdicts** for the factory setting → **VERA-lite.**

The VERA-lite pipeline

1

Guiding questions

Start from concrete yes/no questions, one per anomaly family (count, colour, orientation, cap, belt motion, intra-object motion).

2

Learner VLM

The frozen model watches the video and answers every question YES/NO with the visual evidence, then emits a structured JSON verdict.

3

Optimiser VLM

It reads the learner's mistakes and rewrites the questions — sharper wording, no weights touched. We ran it once (v2 → v3).

4

Blind evaluation

The learner NEVER sees the ground-truth label. An early leak was fixed; the first honest score was 0.77, then 0.92 after v3.

Result of the loop: rewriting the questions from negative constraints to positive verification lifted blind accuracy from **0.77 → 0.92**.

How the guiding questions were hardened (v2 → v3)

The optimiser noticed: **"VLMs struggle with logical negation."** It shifted prompts from negative constraints to positive verification.

Anomaly family	v2 (negated / weak)	v3 (positive / hardened)
Count	no missing pen and no empty gap	**exactly three pens in a single row**
Colour	none differing in colour	**identical barrel & cap colours, no mismatch**
Belt motion	without stopping, pausing, reversing	**advances at every moment, never halting even briefly (1–3 s)**

Positive, concrete phrasing removed the model's blind spots on count, colour and belt-stop cases.

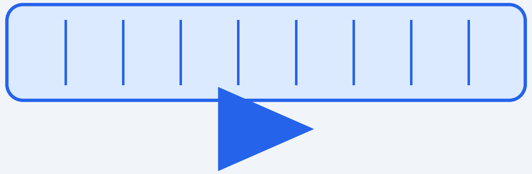
Inside a verdict — structured, auditable JSON

For every clip the model returns structured JSON: a YES/NO answer **with the visual evidence** for each guiding question, then the verdict, type, score and confidence.

On the wrong-colour clip, a single **"NO"** on the colour question drives the decision. The per-question evidence makes the verdict **auditable on the factory floor.**

```
{
  "question_answers": [
    {q1 "YES"  three pens in a row}
    {q2 "YES"  caps aligned, same way}
    {q3 "NO"   leftmost pen is BLUE}
    {q4 "YES"  all capped, similar length}
    {q5 "YES"  board moves, no stop}
    {q6 "YES"  pens fixed to the board}
  ],
  "verdict":      "ANOMALY"
  "anomaly_type": "wrong_color"
  "anomaly_score": 1.0
  "confidence":   "HIGH"
}
```

Three ways we fed the clip to the model

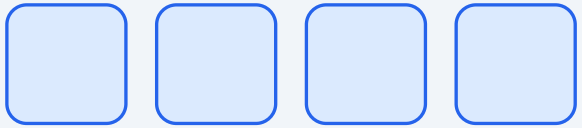


one continuous stream

Continuous video

Send the whole clip; the VLM samples a few frames internally.

12/13 · misses temporal

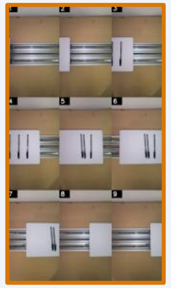


N discrete frames

Dense frames

Extract N evenly-spaced frames, send them as separate images.

11/13 · catches temporal



3×3 montage we actually sent

Temporal grid

Tile N time-ordered frames into ONE montage image, single pass.

negative · ≈ dense + a false positive

The trade-off is intrinsic: continuous (video) catches edges & belt-stop; discrete (frames / grid) catches intra-object motion. We confirmed it on 3 discrete variants — **no single representation reaches 13/13.**

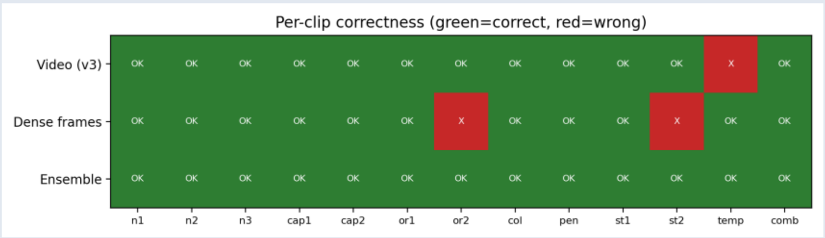
THE SOLUTION

Disjoint blind spots → combine into an ensemble

The hard case (temporal): pens shift mid-clip while the belt moves normally. Full-video input misses it — the model samples only a few internal frames. Dense frames catch it but miss some subtle cases. Their blind spots are **disjoint**.

Input method	Accuracy	False pos.	Temporal?
Video (continuous)	12/13 (0.92)	0	**No**
Dense frames (16f)	11/13 (0.85)	0	**Yes**
OR-ensemble (video ∨ dense)	**13/13 (1.00)**	**0**	**Yes**

Two views, disjoint blind spots → the union (OR) solves every clip with zero false alarms.



Per-clip correctness: only the ensemble is fully green.

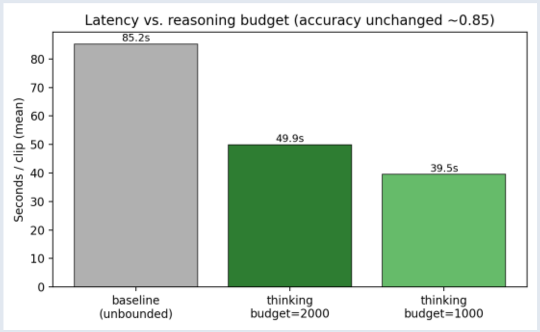
Accuracy, speed and determinism

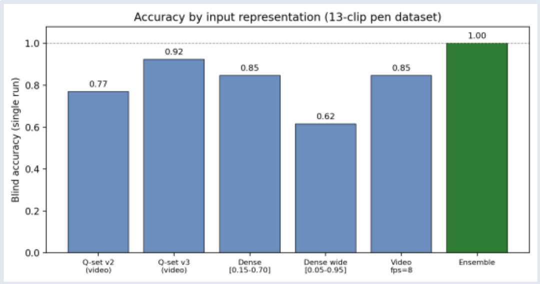
GEBZE TECHNICAL UNIVERSITY

13/13	0	0.846	−41%
Ensemble accuracy	False positives	Repeatable single-view	Latency after tuning

Speed & determinism

- qwen3.6-plus is a reasoning model — it once spent 15,684 reasoning tokens, spiking latency to 85.2 s/clip.
- Capping thinking_budget = 2000 → 49.9 s/clip (−41%).
- **Bonus: the cap made it deterministic. Verdict flips across 3 runs dropped from 2 clips to 0.**
- Honest, repeatable accuracy is 0.846 (11/13 every run) — not the lucky single-run 0.92.





Limitations & open problems

GEBZE TECHNICAL UNIVERSITY

Small, single-scene dataset

13 clips on one rig — results are indicative, not statistically conclusive. A larger, multi-scene set is needed.

LLM non-determinism

The same input could flip the verdict; we needed a thinking-budget cap to stabilise it. True accuracy (0.846) is below the lucky 0.92.

No single view catches all

Continuous video misses intra-object motion; dense frames miss subtle cases. The 13/13 needs an ensemble of two API calls.

Not real-time / cloud-bound

~50 s per clip, depends on a paid API and internet. Raises latency, cost and data-privacy concerns for a real factory.

Limited model comparison

Other models were inaccessible (qwen-vl-max-latest → 403, qwen3.7 → 400); results tie to one vendor model version.

Question tuning sees labels

The optimiser used labels on a tiny set — risk of overfitting the questions to these specific clips.

Explainable VAD without any labelled training data is viable for industry.

GEBZE TECHNICAL UNIVERSITY

WHAT WE SHOWED

- A frozen VLM + learned guiding questions detects fine-grained line anomalies, zero training.
- Every verdict is a human-readable, auditable JSON.
- A videoVdense ensemble reaches 13/13 with 0 false alarms.
- A thinking-budget cap makes it 41% faster and deterministic.

FUTURE WORK

- Cascade: a cheap CV detector first, the VLM only on flagged segments → real-time & cheaper.
- Larger, multi-scene dataset for statistically solid numbers.
- On-prem / open VLM to remove cloud cost and privacy concerns.
- Automate the dense+video ensemble into one call.

References

1. Fujimoto, R. M., “Parallel and Distributed Simulation Systems”, John Wiley & Sons Inc, 2000
2. Mattern, F., “Efficient Algorithms for Distributed Snapshots and Global Virtual Time Approximation”, Parallel and Distributed Computing, Vol. 18 No. 4, 1993
3. Bauer, D., Yaun, G., Carothers, C., Yuksel, M., Kalyanaraman, S., “Seven-O'Clock: A New Distributed GVT Algorithm Using Network Atomic Operations”, PADS 2005
4. MSDN , developer network [online], http://msdn.microsoft.com/en-us/library/dd409390.aspx [Ziyaret Tarihi: 9 Eylül 2013].